242405: metamonzogranite, Mount Gibson mine

(Thundelarra Supersuite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Ninghan (SH 50-7), Mount Gibson (2338)
MGA Zone 50, 513503E 6707680N

Sampled on 12 May 2019

This sample was collected from outcrop on Mount Gibson Station, west of the Mount Gibson mining area, about 9.9 km east of the Goodlands Road intersection with the Great Northern Highway, 1.9 km west of the Wombat open pit, and 1.0 km south of the Mount Gibson mine accommodation village.

Geological context

The unit sampled is a metamonzogranite of the Thundelarra Supersuite. The 3010–2930 Ma Thundelarra Supersuite consists of metagranitic to metatonalitic rocks, and is the oldest basement unit in the Murchison Domain, coeval with the oldest recognized felsic metavolcanic rocks, represented by the Madoonga, Gossan Hill, and Mount Gibson Formations (Van Kranendonk et al., 2013). A drillcore sample of Mount Gibson Formation metadacite, from a drillcore about 23 km to the north, yielded an igneous crystallization age of 2932 ± 5 Ma (GSWA 242407, Wingate et al., 2021). A metamonzogranite of the Thundelarra Supersuite, sampled about 101 km to the northwest, yielded an igneous crystallization age of 2946 ± 4 Ma (GSWA 219311, Wingate et al., 2020).

Petrographic description

The sample is medium-grained metamonzogranite (Fig. 1), consisting of about 35% plagioclase, 30% quartz, 20–25% K-feldspar, 7–8% biotite, 3% myrmekite, and trace magnetite, hematite, muscovite, apatite and zircon. The rock was probably originally equigranular, but is partially recrystallized, forming fine-grained patches of polygonal or anhedral grains. Plagioclase is anhedral to subhedral, equant to tabular, and up to 2.5 mm long; within the recrystallized patches, plagioclase is xenoblastic and up to 0.4 mm in diameter. Quartz forms irregular, monomineralic patches up to 12 mm in diameter, containing anhedral intergrown grains that exhibit undulose extinction and are up to 3 mm in size. Quartz forms xenoblastic polygonal grains in the recrystallized patches. Microperthitic K-feldspar is anhedral, equant, and up to 5 mm in size, and forms polygonal grains up to 0.4 mm diameter in the recrystallized patches. Some K-feldspar displays microcline twinning. Rounded lobes of myrmekite up to 1 mm in size occur on the margins of some of the larger K-feldspar grains. Biotite forms anhedral to subhedral laths up to 2 mm long that are partly overprinted by aggregates of randomly oriented subhedral laths up to 0.5 mm long. Muscovite forms subhedral laths up to 0.5 mm long, and subhedral magnetite grains are up to 0.4 mm in diameter. Bladed apatite grains are up to 0.1 mm long. Feldspars are weakly altered by sericite and clay minerals, and biotite shows minor chlorite alteration. The sample lacks a foliation although microcline twinning and undulose extinction are evidence of deformation and grain boundary adjustment. The recrystallized patches of fine-grained feldspar and quartz and development of myrmekite suggest metamorphism at relatively high-temperature amphibolite conditions.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, and subhedral to euhedral. The crystals are up to 350 µm long, and elongate, with aspect ratios up to 7:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and many crystals are dominated by high-U, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 4–5 September 2020, using SHRIMP-B. Nine analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.21% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.55%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Sixteen analyses were obtained from 16 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Figure 1. Hand specimen image of sample 242405: metamonzogranite, Mount Gibson mine

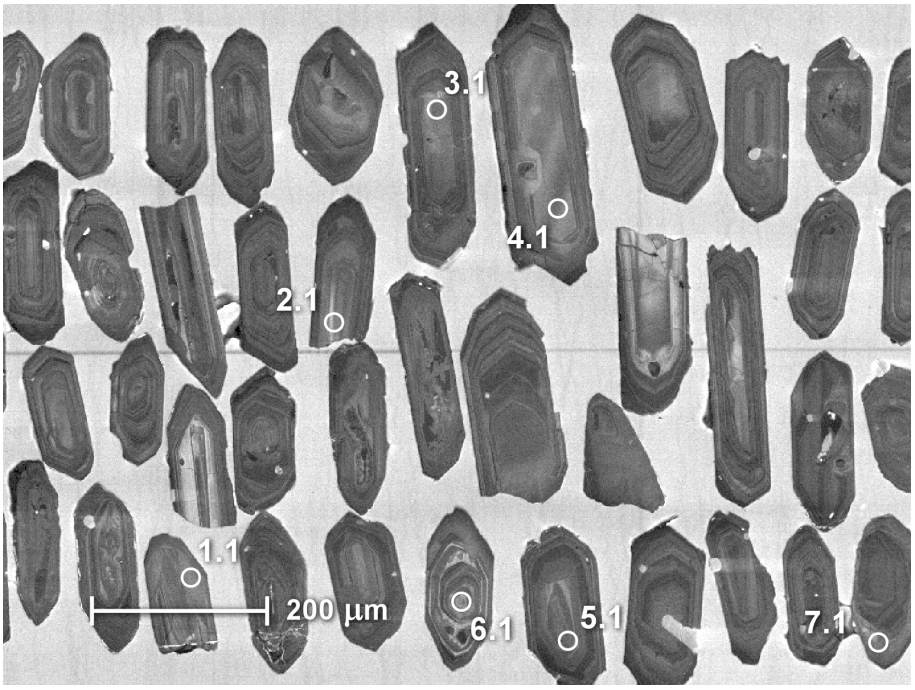

Figure 2. Cathodoluminescence image of representative zircons from sample 242405: metamonzogranite, Mount Gibson mine. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 242405: metamonzogranite, Mount Gibson mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	1	1.1	148	29	0.20	0.050	1.753	0.019	0.21185	0.00129	1.753	0.019	0.21140	0.00131	2909	26	2916	10	0.2
I	11	11.1	214	81	0.38	0.018	1.760	0.017	0.21171	0.00114	1.760	0.017	0.21155	0.00115	2901	23	2917	9	0.6
I	8	8.1	344	150	0.44	0.060	1.774	0.020	0.21282	0.00100	1.775	0.020	0.21230	0.00102	2881	27	2923	8	1.4
I	2	2.1	137	65	0.48	0.083	1.765	0.020	0.21303	0.00133	1.767	0.020	0.21231	0.00137	2891	27	2923	10	1.1
I	4	4.1	293	111	0.38	0.007	1.764	0.016	0.21244	0.00105	1.764	0.016	0.21238	0.00105	2895	21	2924	8	1.0
I	3	3.1	194	69	0.36	0.108	1.963	0.020	0.21350	0.00122	1.965	0.020	0.21254	0.00126	2652	22	2925	10	9.3
I	13	13.1	225	119	0.53	-0.017	1.775	0.017	0.21261	0.00114	1.775	0.017	0.21276	0.00114	2881	23	2927	9	1.6
I	5	5.1	477	325	0.68	0.139	1.831	0.014	0.21402	0.00093	1.834	0.014	0.21279	0.00096	2806	18	2927	7	4.1
I	15	15.1	306	349	1.14	0.039	1.785	0.016	0.21314	0.00105	1.786	0.016	0.21279	0.00106	2867	21	2927	8	2.0
I	16	16.1	405	176	0.43	0.030	1.839	0.015	0.21391	0.00138	1.839	0.015	0.21364	0.00138	2799	19	2933	10	4.6
I	7	7.1	176	112	0.64	0.259	2.246	0.023	0.21596	0.00129	2.252	0.023	0.21367	0.00138	2369	20	2934	10	19.2
I	6	6.1	185	95	0.51	0.037	1.986	0.030	0.21421	0.00128	1.986	0.030	0.21389	0.00130	2629	33	2935	10	10.4
I	10	10.1	267	178	0.67	-0.007	1.954	0.018	0.21390	0.00182	1.954	0.018	0.21396	0.00182	2664	20	2936	14	9.2
I	14	14.1	186	73	0.39	0.011	1.789	0.018	0.21473	0.00125	1.790	0.018	0.21463	0.00126	2862	24	2941	9	2.7
I	12	12.1	166	96	0.58	0.034	1.750	0.018	0.21508	0.00125	1.750	0.018	0.21478	0.00126	2914	25	2942	10	1.0
D	9	9.1	125	178	1.43	0.513	3.212	0.066	0.21892	0.00337	3.229	0.067	0.21438	0.00356	1739	32	2939	27	40.8

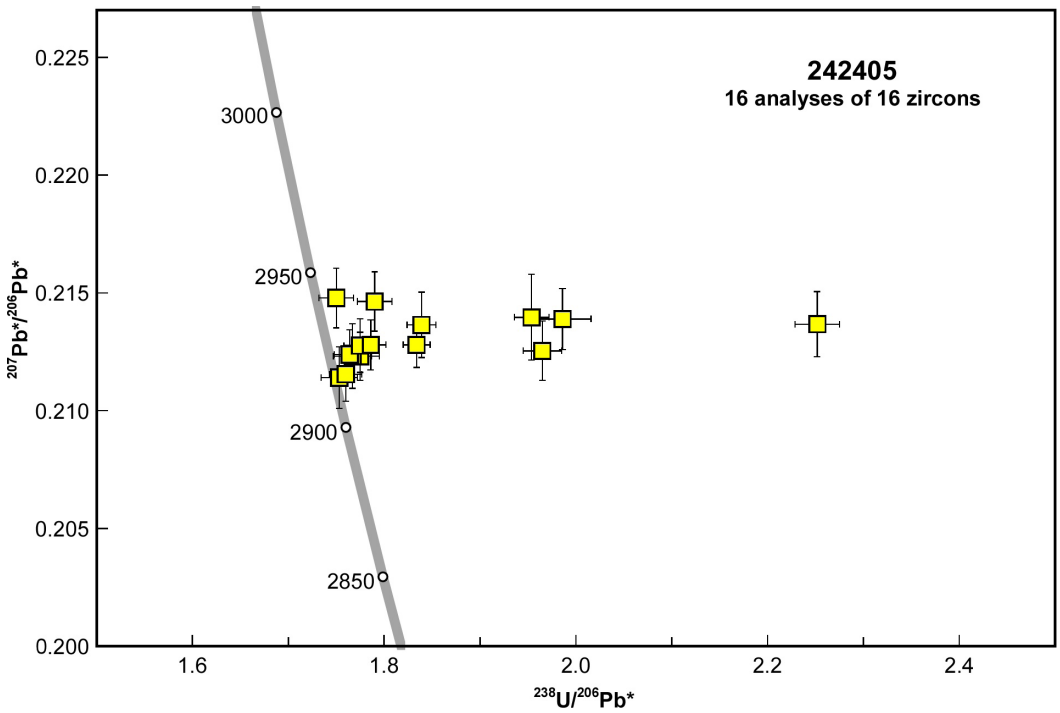

Figure 3. U–Pb analytical data for sample 242405: metamonzogranite, Mount Gibson mine. Yellow squares indicate Group I (magmatic zircons). One highly discordant analysis in Group D is not shown

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). One analysis is >40% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and considered not to be geologically significant. The remaining 15 analyses form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 15 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2928 ± 5 Ma (MSWD = 0.66).

The date of 2928 ± 5 Ma for the 15 analyses in Group I is interpreted as the igneous crystallization age of the monzogranite.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, T 2021, 242405: metamonzogranite, Mount Gibson mine; Geochronology Record 1805: Geological Survey of Western Australia, 4p.

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Data released: 21 October 2021